3.5-3.11 Weekly Record

Continue processing `attributes`

Removed manually unwanted attributes: {'WiFi', 'HasTV', 'BusinessAcceptsCreditCards'}

Formatted the `Ambience` attribute from a nested dictionary structure (e.g., `{'touristy': False, 'hipster': False, 'romantic': False, 'divey': False, 'intimate': False, 'trendy': False, 'upscale': False, 'classy': True, 'casual': False}`) into a **string** representation, facilitating smooth merging with the text description in subsequent processing steps.

Attribute Analysis

- **Ambience**: Too complex for structured format; merged into textual description.
- **RestaurantsPriceRange2**: **Ordinal Variable** ranging from 1 to 4. To avoid increasing dimensionality from one-hot encoding, being normalized (?) and integrated into the item's description.
  - *RestaurantsPriceRange2 Interpretation:*

    Higher numbers indicate higher prices.

    https://www.yelp.com/topic/san-diego-can-anyone-give-me-the-actual-dollar-range-for-the-dollar-sign-symbols-in-rrgards-to-pricing

    - **$**: Under $10
    - **$$**: $11–$30
    - **$$$**: $31–$60
    - **$$$$**: Above $61
- **RestaurantsDelivery, OutdoorSeating, BusinessParking, BikeParking, RestaurantsTakeOut, RestaurantsReservations**: Binary (True/False); directly binarized for structured representation.

Preparation: Construct Description

Transform Attributes:

- `RestaurantsPriceRange2`: Converted numerical ranges (1–4) to descriptive text, omit if unknown;
- `Ambience`: Converted structured attributes into descriptive text; omit if unknown.

`categories` **Filtering & Clustering**:

- Assign businesses to clusters from hybrid clustering results.
- Only retain those belong to 195 valid categories and merge them into description.

Construct Descriptions:

- Dynamically combine attributes into textual descriptions. (`RestaurantsPriceRange2`, `Ambience`, `Categories`)

"description": "La Tartine: categorised under Creperies, Mediterranean, Coffee & Tea, Juice Bars & Smoothies",

"description": "Sultan Indian Cuisine: a moderately priced restaurant with casual, classy ambience categorised under Pakistani, Kosher, Indian"

Conduct statistical analysis based on `pa_valid_filtered_dining_4.json`, generated from the operations described above.

Statistical Analysis:

- Counted categorical/binary attribute occurrences.
- Summarized categorical attribute uniqueness.
- Computed descriptive statistics (mean, median, min, max, std) for numerical attributes.

Key Findings:

- `stars`: Normally distributed (3.0–5.0).
- `review_count`: Skewed (max: 4,250; mean: 117.86; median: 54.5).

- `RestaurantsPriceRange2`: 15.7% missing values.
- Binary attributes: Contain `"Unknown"`; conversion to binary needed.
- Category clusters: Imbalanced but realistic.

```
"category_cluster": {
    "7": 1345, // take-away, dessert, drinks
    "5": 491, // pub
    "6": 956, // street food
    "2": 717, // Asian
    "4": 2542, // western classics
    "3": 298, // global
    "1": 267 // latin American
  }
```

Data Preprocessing/Item Profile Building Step 1

Impute Missing Values:

- Filled missing `RestaurantsPriceRange2` using median values by `category_cluster`.

Normalize Attributes:

- Scaled `RestaurantsPriceRange2` into a normalized range (0–1).

Binary Attribute Conversion:

- Transformed binary attributes (`True` → 1, `False` → 0).
- Handled `Unknown` values by treating them as `False` (0).

Compute Quality-Popularity Score (for future similarity calculation):

- Applied Bayesian smoothing to derive `qual_pop_wt_score` (This method adjusts the scores by incorporating a global average rating, effectively balancing items with few reviews against items with many reviews.).

$$\text{qual\_pop\_wt\_score} = \frac{\text{stars} \times \text{review\_count} + m \times \bar{s}}{\text{review\_count} + m}$$

Global Average Rating: 3.8511
Smoothing Factor (m):
`m ≈ median(df["review_count"])` (to be modified)

Improvements:

1. Add a `sentiment_score` to enhance the quality assessment.

   Proposed Approach: Start by selecting **1-2 restaurants** as test cases; randomly sample and analyze their **associated reviews.**

2. Current rating is skewed due to filtering (`rating ≥ 3`, `review_count ≥ 10`).

   Recalculate the **Global Average Rating** using the dataset that are **before applying** the filtering conditions (`rating ≥ 3`, `review_count ≥ 10`).

Data Cleaning:

- Removed unnecessary columns, retaining only relevant fields.

Next Steps

Drafting: Write completed data processing.

Structure the writing, like a tutorial explaining to someone new to the research (or to my past self): for each part - what was done, why, and how.

Profile Creation: Build comprehensive item profiles and user profiles.

- **Score Optimization**: Tune *m* or explore alternative scores.
- **Review Integration**: Link to review and tip to enrich item profiles (Q: How many reviews/tips per item, how to choose, and integrate into description or separate?) 10 reviews; the most recent ones; merged with tips, and keep a separate new field “review_tip”

{"review_id":"KU_O5udG6zpxOg-VcAEodg",
"user_id":"mh_-eMZ6K5RLWhZyISBhwA",
"business_id":"XQfwVwDr-v0ZS3_CbbE5Xw",
"stars":3.0,
"useful":0,"funny":0,"cool":0,
"text":"If you decide to eat here, just be aware it is going to take about 2 hours fr
"date":"2018-07-07 22:09:11"}

{"user_id":"OttfcRxgRrYsTg9EV5Aozg",
"business_id":"clwjLY7PdYJpe7IP9lrqEw",
"text":"Order the Tortilla Soup",
"date":"2014-06-17 01:20:14",
"compliment_count":0}